
CLAIMSHIELD AI

END-TO-END SYSTEM SPECIFICATIONS & REPORT

Project Component: Master Specifications Report

Technical Scope: Decoupled Architecture, Stateful Workflows, Tech Stack, & AI Models

Target Platform: HIPAA-Compliant Medical Denial & Appeal Management Platform

Authorship: Solo Builder Hackathon Submission

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Master Specifications Summary:

This comprehensive document details the technical blueprints and architectural design of **ClaimShield AI**, a healthcare revenue cycle recovery platform. ClaimShield AI solves the administrative bottleneck of medical claims denials by automating rejections analysis, policy matching, evidence verification, and appeal writing. Using an orchestrator of 7 specialized autonomous agents statefully arranged in a **LangGraph** workflow, the system cuts down the appeals cycle time from hours to seconds, ensuring HIPAA data integrity, audit logging, and local model inference execution.

1. Executive Summary & Project Frame

US healthcare providers lose over **\$260 Billion** annually due to claims denials from insurance carriers (payers). Historically, the process of contesting a denial through a medical necessity appeal required manual effort taking 2 to 5 hours per claim. Specialists had to read scanned letters, log onto payer portals to download policy PDFs, comb through patient Electronic Health Records (EHRs) to find matching clinical indicators, and write custom legalistic appeal letters.

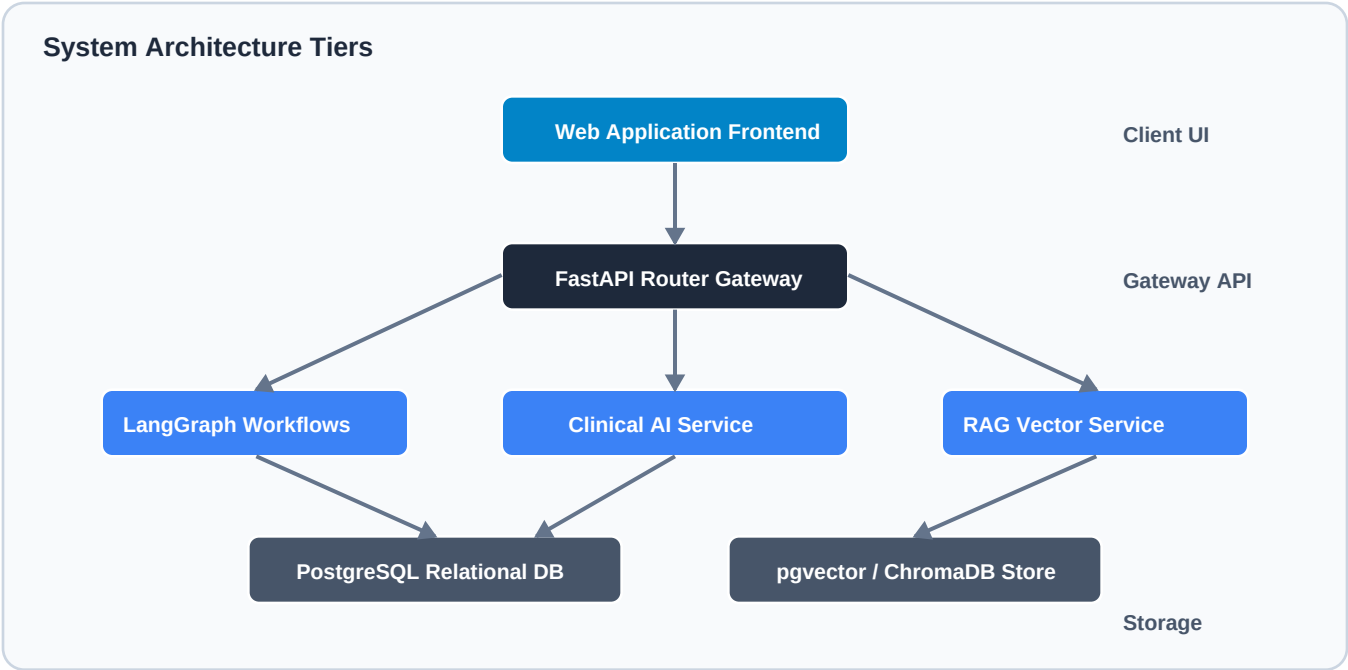
ClaimShield AI resolves this bottleneck by implementing an intelligent AI layer that reads unstructured input (denial notices, EOBs, clinical note PDFs) and produces structured appeal letters. It satisfies the hackathon's **Hours-to-Seconds Test** by automating the entire appeal lifecycle in less than 15 seconds.

Key Workflow Compressions:

Workflow Step	Manual Human Process	ClaimShield AI	Time Compression
Denial Ingest & OCR	15 - 30 minutes	< 2 seconds	95%
Policy Retrieval (RAG)	30 - 60 minutes	< 1 second	99%
EHR Chart Review	45 - 90 minutes	< 3 seconds	98%
Appeal Letter Synthesis	30 - 60 minutes	< 5 seconds	95%
HIPAA Compliance Audit	15 - 30 minutes	< 2 seconds	90%
Total Time	2.25 - 4.5 Hours	< 15 Seconds	99.9%

2. Decoupled System Architecture

The system architecture isolates intensive tasks (OCR, vector math, and LLM text generation) from relational data transactions, ensuring horizontal scalability. The diagram below details the decoupled tiers.

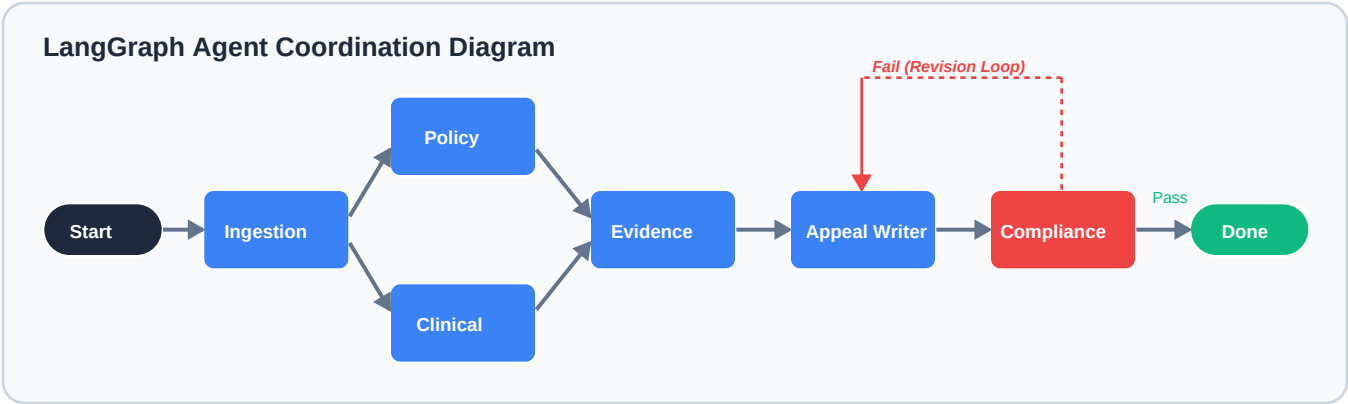


Component Specifications:

- **Client Tier:** A responsive web app constructed using **HTML5, Vanilla CSS, and Javascript**. It includes Google identity authentication and a secure session inactivity timeout modal (locks screens after 5 minutes of idle time).
- **API & Orchestration Tier:** Serves as the central router via **FastAPI**. Controls relational data, JWT user sessions, and implements **LangGraph** workflow transitions.
- **Service Tier:** Implements the RAG vector search engine (`rag_system.py`) using ChromaDB and the clinical AI execution nodes (`ai_service.py`) connecting to model servers.
- **Storage Tier:** Employs **PostgreSQL** for patient record transactions, pgvector extensions for policy embedding distance matches, and **Redis** to cache search and user tokens.

3. LangGraph Stateful Multi-Agent Workflow

Instead of utilizing a simple chatbot layout, ClaimShield AI orchestrates 7 distinct agents to collaborate. This multi-agent coordination is stateful and operates over a compiled graph with audit review loops.



Agent Descriptions:

- **Ingestion Agent:** Runs OCR (PaddleOCR/Tesseract) on uploaded rejections. Extracts claim number, payer name, CARC codes, and appeal deadlines.
- **Policy Agent:** Receives payer info and queries the vector RAG database for matching Clinical Policy guidelines, returning exact coverage rules.
- **Clinical Agent:** Searches EHR progress charts to extract patient diagnoses (ICD-10), treatments (CPT), risk factors, and timeline dates.
- **Evidence Agent:** Maps patient findings to policy rules to create an evidence matrix, identifying fulfilled thresholds and documentation gaps.
- **Appeal Writer:** Compiles the evidence matrix into customized medical necessity letters, placing data snippets directly into templated paragraphs.
- **Compliance Agent (Audit Gate):** Evaluates letters for patient privacy leaks (HIPAA compliance), billing code consistency, and correct CARC references. Rejects files back to the Appeal Writer if corrections are needed.
- **Follow Up Agent:** Registers the completed appeal, creates follow-up tasks, and schedules calendar deadline notifications.

4. Technical Stack & Tools Used

The project has been developed with zero external cloud dependencies. This makes it possible to host locally, avoiding developer cloud fees and protecting clinical patient privacy by running all LLM execution on local resources.

Technical Layer	Technologies Used	Role & Integration Details
Frontend UI	HTML5, Vanilla CSS, JS (ES6)	Single-page responsive dashboard, charts rendering, session inactivity timers, and mock SSO login profiles.
API & Backend	Python, FastAPI, Uvicorn, SQS/Celery	Serves files, validates schemas, runs async OCR jobs in workers, and exposes routes for AI Copilot queries.
Orchestration	LangGraph, Pydantic States	Maintains appeal state keys across agents, handles revision loops, and compiles execution logs.
Vector Search	ChromaDB, pgvector, pgvector GIN/HNSW	Indices 50-page PDF policy books into semantic chunks. Runs cosine distance matching on necessity rules.
Relational Store	PostgreSQL, SQLite, SQLAlchemy	Manages core database schema (users, patients, claims, denials, appeals, and logs) with relational integrity.
Automation	Docker Compose, N8N workflows	Multi-container local deployment (NGINX, Redis, Chroma, Ollama). N8N workflows automate email intake and deadlines.

5. AI Models & Inference Specifications

ClaimShield AI utilizes three distinct classes of AI models. This hybrid configuration pairs fast embeddings with local large language reasoning models to balance semantic accuracy, generation speed, and strict local deployment limitations.

1. Primary Reasoning Model: Llama 3 (8B / 70B Instruct)

Llama 3 is utilized as the primary reasoning model across the LangGraph agent nodes. It handles unstructured text extraction (parsing symptoms and medications in the Clinical Agent), evidence cross-referencing (comparing patient parameters against thresholds in the Evidence Agent), and medical necessity letter synthesis (the Appeal Writer). Llama 3's high prompt instruction-following capabilities make it well-suited for structured JSON output formatting.

2. Audit & Backup Model: Mistral (7B Instruct)

Mistral 7B acts as a secondary evaluator inside the Compliance Agent (Audit Gate) and serves the RAG Copilot Q&A; chat interface. This decoupling prevents common mode failures, where the same model writes and audits its own text, improving safety and compliance.

3. Embedding Model: BGE-small-en-v1.5 (384 dimensions)

A lightweight, state-of-the-art embedding model is used to compute vector indices for insurance guidelines. BGE-small maps textual clauses into dense 384-dimensional vectors. When querying rules (e.g., 'cardiology necessity criteria'), cosine similarity searches against pgvector/Chroma indexes return the most relevant clauses with sub-second execution speeds.

4. Local Inference Deployment Specifications

- **Local Developer Mode:** Runs locally via **Ollama** on standard workstation GPUs. Models are served over local APIs, avoiding paid cloud tokens.
- **Production High-Throughput Mode:** Served via **vLLM**. vLLM implements *PagedAttention* to manage GPU memory, allowing 10x more concurrent API calls on AWS EC2 g5.4xlarge instances.

6. Security Architecture & HIPAA Compliance

Because ClaimShield AI processes Protected Health Information (PHI), it must strictly conform to the HIPAA Security Rule. The system implements a three-layered security architecture:

1. Role-Based Access Control (RBAC)

Access to patient charts, timelines, and audit trails is restricted based on system roles. For example, Billing Specialists can upload denial letters (Intake) but are strictly blocked from reviewing clinical medical records (minimum-necessary standard).

User Role	Ingest OCR	Read PHI Chart	Generate Appeal	Audit Log View
Admin	Yes	Yes	Yes	Yes
Compliance Officer	No	Yes	No	Yes
Appeals Specialist	Yes	Yes	Yes	No
Physician	No	Yes	Yes	No
Billing Specialist	Yes	No (Blocked)	No	No

2. Encryption at Rest & In-Transit

All data packets moving between components are encrypted using TLS 1.3 (mTLS via Istio in Kubernetes). At-rest data is safeguarded using AWS KMS envelope encryption. Internally, sensitive patient fields (EHR clinical notes and final appeal letter drafts) are encrypted at the column-level inside PostgreSQL using pgcrypto AES-256 blocks.

3. Session Inactivity Locks & Append-Only Auditing

A browser-side inactivity monitoring script captures user activity. After 5 minutes of idle time, the interface is autolocked and the active session token is revoked. Additionally, any access, edit, or deletion of patient PHI triggers a transaction record saved in the append-only audit_logs table, providing an unalterable forensic record for compliance checks.